

returns of gold with lagged twenty-year real returns of gold is -63% —a staggering number in absolute value terms. When the real price of gold has been high, the subsequent long-term performance of gold has been low. This should be a warning to all asset owners adding to their gold holdings now. The classic error of an investor chasing returns is to invest after past returns have been high, while current prices are high. Subsequent returns often end up disappointingly low.

I close with a look at the real performance of gold relative to equities in Panel B of Figure 11.10. The line for gold is exactly the same as in Panel A, except the scale on the y-axis has been drawn to be logarithmic. Panel B overlays the real price of equities also starting at 1.0 at September 1875. The equity market simply clobbers gold in real terms.

6. Real Estate

Real estate is all about space. To be more specific, real estate investing is about creating and maintaining structures to remove physical space between people and institutions. There are two core models in urban economics that articulate the fundamental sources of demand and supply for real estate. Central to both is the concept of *spatial equilibrium*, which states that people and firms invest in real estate in such a way that they are indifferent about where they locate.²⁸ Agents' utility and, ultimately, real estate values, are affected by the amenities provided by real estate—which include proximity to customers and suppliers, human capital spillovers, and the innate advantages of geography, including gorgeous views and nice weather. Real estate values are also affected by workers' income, housing costs, and transportation costs.

The Alonso (1964), Mills (1967), and Muth (1969) model explains real estate prices *within* a given city in terms of distance to centers of production: as transportation costs increase, real estate prices fall. The second workhorse model, Rosen (1979) and Roback (1982), explains how real estate prices differ *across* different cities and metropolitan areas because of different incomes, amenities, and productivity levels. Higher land values are the price of entry to metropolitan areas with more productive firms or better amenities (which includes more cultural activities, less crime, better schools, and more comfortable weather).

The realtors' mantra of "location, location, location" gets it only half right. Location certainly is the primary driver in the Alonso-Mills-Muth model, but it matters because of transportation costs. The Rosen-Roback model emphasizes that it is not location per se but the interaction of location's advantages (enabling trade or production, for example) with the productive capabilities of firms and workers and other macro factors.

²⁸ See Glaeser (2008) for a summary.